TECHNICAL PRESENTATION

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Introduction → Education → Career → Technical Presentation → Q&A

Role: Engineer

Focus: Distributed Systems

Hardware foundations. Software ambitions.

Anna University
B.E. Electronics & Communication Engineering
Microprocessors, Embedded Systems, Digital Electronics

Northeastern University
M.S. Computer Software Engineering
Algorithms, Database Design, Cloud, DevOps

COMPANY	ROLE	HIGHLIGHTS
Cognizant	Project Lead	Mainframe → ETL migration. 4 teams, delivered through COVID.
Digital HIE	Tech Lead	HIPAA accreditation, solo-built compliance infrastructure.
Capital One	Systems Architect	130M+ accounts migrated to serverless AWS architecture.

01 — PROBLEM STATEMENT

One monolith. Four LOBs. It couldn't scale.

BEFORE	
Architecture	Monolith
Coupling	Tight
Deployment	All-or-nothing
Scaling	Vertical only
Risk	Shared failures

AFTER	
Architecture	Microservices
Coupling	Loose
Deployment	Independent
Scaling	Horizontal
Risk	Isolated

Monolith → C Card Svc | B Bank Svc | A Auto Svc | G Global Svc → Lib Shared Libs | API Reusable APIs → AWS Cloud Infra

02 — METHODOLOGY & ARCHITECTURE

Let the product decide the stack.

PHASE	DECISION
Requirements	130M+ accounts, 4 LOBs
Schema	NoSQL single-table design
Database	DynamoDB
Language	Go (concurrency + cold start)
Cloud	AWS Lambda + SQS + ALB

REQUEST FLOW

Automated pipeline. 98% coverage. Zero manual gates.

DEPLOYMENT PIPELINE

TEST TYPE	RESULT
SonarQube	98% Quality Gate
Unit Tests	100% Coverage
Integration	BDD / Gherkin
Performance	1B+ Records

BDD-first approach: Business stakeholders wrote Gherkin specs before engineering started. Every feature traced back to an acceptance criterion. Zero ambiguity at handoff.

Performance validated against 1B+ synthetic records across all four LOBs before production cutover.

Challenge existing designs. Let the product decide.

SERVICE ARCHITECTURE

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Core Services

↓ ↓ ↓ ↓

B	C	A	G
Bank DB	Card DB	Auto DB	Global DB

ACCOUNTS	DELIVERED	SERVICES	COVERAGE
130M+	8 mo	4	98%
All LOBs	End-to-end	Code-once	Sonar

- **Product requirements define architecture.** Schema → DB → Cloud → Language.
- **Challenge inherited designs.** Better approaches hide behind assumptions.
- **Root-cause analysis over gut.** Volumes + growth rate = foundation for every decision.
- **Ripple effect.** Other teams adopted the patterns including batch processing.

> *"Terraform-managed cross-region failover gave us active-active resilience without manual intervention. One config, two regions, zero downtime."*

PERSONAL & Q&A

Beyond the code.

🏏 **Sports**
Volleyball, Cricket, Badminton.

🚗 **Driving**
Manhattan & Jersey City explorer.

🌐 **Languages**
English, Telugu, Tamil, Hindi, Kannada.

🍽️ **Exploring**
New cuisines, new places, new conversations.

Ok, shoot.

Happy to go deeper on any decisions, trade-offs, or challenges.

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